

Eshan Mohamed Jaffar

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• <https://eshanjaffar.github.io/>

Education

Chennai Mathematical Institute

August 2023 – April 2026

Degree

Major Mathematics

Minor Computer Science

Courses : Probabilistic Number theory, Analytic Number theory, Diophantine Approximation, Rudiments of Riemann zeta function, Zeta functions of Number fields

Projects

Modular Forms

February 2026 - Present

Prof. B. Ramakrishnan

- Studied the action of $SL_2(\mathbb{Z})$ on the upper half-plane and congruence subgroups $\Gamma(N)$, $\Gamma_0(N)$, $\Gamma_1(N)$
- Learned modular forms for $SL_2(\mathbb{Z})$, including transformation laws, q -expansions, and the valence formula
- Studied Eisenstein series $E_k(z)$ and their Fourier expansions via divisor sums
- Explored the discriminant form $\Delta(z)$

Apollonian Circle packings

October 2025 - Present

Prof. Anwesh Ray

- Studied the algebraic and geometric structure of Apollonian circle packings
- Worked through foundational [paper](#) by Graham et al., followed by a study of Elena Fuchs's [thesis](#)
- Analyzed the associated quadratic forms and the Apollonian group
- Explored arithmetic properties and local-global behavior using tools from algebra and number theory

Measure Theoretic Probability

May 2024 - July 2024

Prof. Rajeeva Karandikar

- Studied measure theory (σ -algebras, measurable functions, Carathéodory extension) to formalize probability spaces
- Proved fundamental convergence theorems (Monotone Convergence Theorem, Dominated Convergence Theorem)
- Reviewed the following [notes](#), followed by the first 2 chapters of “Measure, Integration & Real Analysis” by Sheldon Axler and then the first chapter of “Probability: theory and examples” by Rick Durrett

Seminar Talks

- Introduction to Selberg sieve and Brun–Titchmarsh theorem as part of CMI Student Seminar
- Apollonian circle packings (following Section 2 of Fuchs’ thesis) as part of Analytic Number Theory course
- Real zeros of L-functions and class numbers of quadratic fields (after Mahler) as part of Zeta Functions of Number Fields course
- Moments of the Riemann zeta function (after Kannan Soundararajan) as part of Rudiments of Zeta Function course

Awards & Honors

Homi Bhabha Centre For Science Education (HBCSE)

- Among top **6** in the state of Kerala selected for appearing for Indian National Mathematics Olympiad **2022**
- Among top **30** in India selected as an INMO Awardee in the Indian National Mathematics Olympiad **2022**

Chennai Mathematical Institute (CMI)

- **Shriram Scholarship:** Tuition fee waiver and monthly stipend for undergraduate studies at CMI. **2023**
- Among top **30** high scorers in the Final round of Scholastic Test of Excellence in Mathematical Sciences **2021**

Volunteering Activities

INMOTC Tutor

May 2025

- Delivered a problem-solving lecture for the Indian National Mathematical Olympiad Training Camp on Number Theory; prepared and circulated detailed [notes](#).

IMOTC Grader

May 2025

- Graded team selection test(TST) problems for national training camp(IMOTC) participants to select the students to represent India at IMO and took part in arbitration discussions to ensure fair and consistent evaluation.

Head of Finance, Tessellate College Fest

October 2024 - February 2025

- Led the finance team for the college fest, managing budgets, coordinating financial planning, and ensuring smooth execution of all monetary aspects, thereby honing leadership and organizational skills

Workshops and Conferences

- Comparative Prime Number Theory Symposium, University of British Columbia
- Sieve Methods and Applications, Kerala School of Mathematics

July 2024

June 2025

Technical Skills & Interests

- **Programming & Tools:** Proficient in C++, Python, and Excel; experienced in mathematical computing with NumPy and automation via Unix/Linux shell scripting.
- **Mathematics:** Passionate about applying analytic techniques in number theory.